

# Improving Uncertainty Communication in Scientific Visualization

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## Abstract

Scientific visualization often communicates point estimates while under-communicating uncertainty, which can lead to overconfident or miscalibrated decisions. This proposal investigates how different uncertainty encodings influence accuracy, confidence calibration, and task time. Using a publicly available dataset, we present preliminary descriptive plots and a summary table based on bootstrap confidence intervals to motivate the experimental design. The proposed study compares error bars, distributional summaries, and hypothetical outcome plots (HOPs) in a within-subjects evaluation.

## 1 Introduction

Scientific visualization routinely presents results derived from measurement, modeling, or sampling, each of which introduces uncertainty. When uncertainty is omitted or poorly conveyed, decision makers may draw overly confident conclusions. This project focuses on scientific visualization within the graphics and visualization research area, emphasizing how different uncertainty encodings affect understanding and decision quality. The goal is to identify practical visualization choices that better support accurate interpretation and calibrated confidence in scientific contexts.

## 2 Research Question

Which uncertainty visualization encoding best improves users' decision accuracy and confidence calibration in scientific visualization tasks?

## 3 Hypothesis Statement

Compared to traditional error bars, uncertainty encodings that show distributional variation (e.g., hypothetical outcome plots or distribution summaries) will produce higher decision accuracy and better confidence calibration without increasing task time.

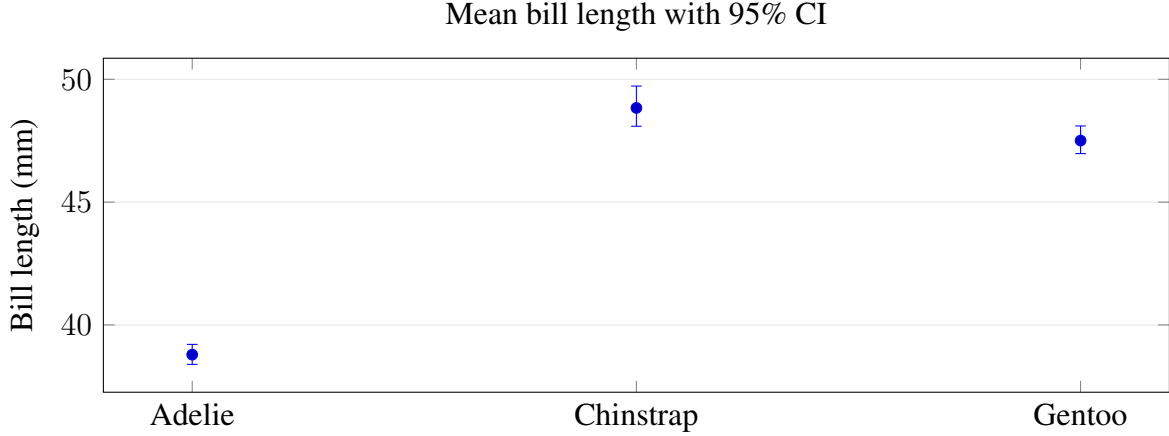


Figure 1: Error-bar summary for bill length by species (Palmer Penguins).

## 4 Preliminary Research Design

- **Overview:** Conduct a within-subjects user study comparing three uncertainty encodings in a scientific visualization task: error bars (baseline), distributional summaries (box/violin), and HOPs.
- **Artifact to Build/Measure:** A visualization dashboard presenting the same datasets across encodings with standardized prompts (ranking, estimation, and decision selection).
- **Baseline:** Error bars representing mean and 95% confidence intervals.
- **Experimental Tasks:** Participants interpret comparisons across groups, choose the most reliable trend, and report confidence.
- **Metrics of Success:** Decision accuracy, confidence calibration (alignment of confidence with correctness), response time, and perceived clarity/trust.
- **Analysis Plan:** Repeated-measures ANOVA or non-parametric equivalents; calibration assessed using Brier score or calibration curves.

## 5 Preliminary Evidence from a Public Dataset

To ground the proposal in real data, we used the Palmer Penguins dataset (openly available on GitHub) and computed descriptive statistics for bill length with bootstrap confidence intervals. These plots illustrate differences between mean-based and distribution-based uncertainty summaries.

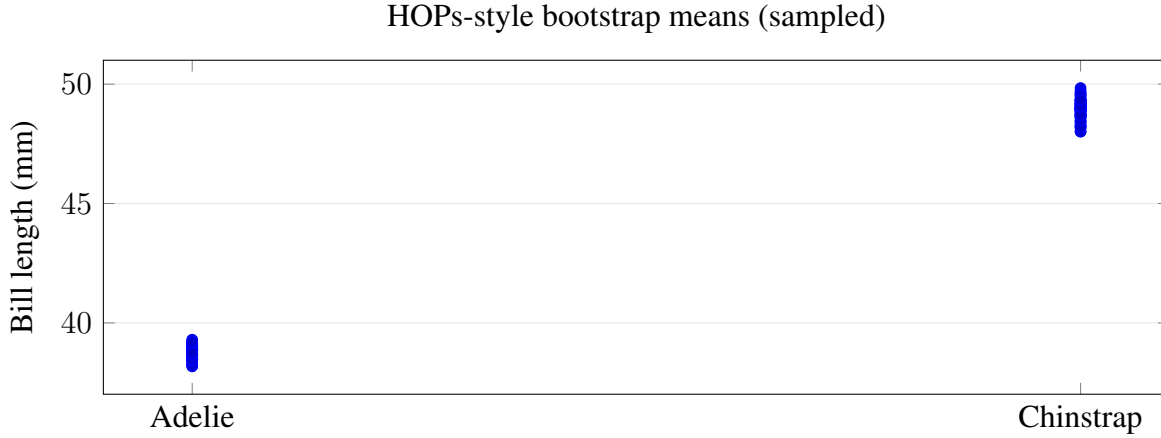


Figure 2: HOPs-style visualization using bootstrap sample means for two species.

| Species   | N   | Mean (mm) | SD (mm) | 95% CI Low | 95% CI High |
|-----------|-----|-----------|---------|------------|-------------|
| Adelie    | 151 | 38.79     | 2.66    | 38.39      | 39.21       |
| Chinstrap | 68  | 48.83     | 3.34    | 48.09      | 49.72       |
| Gentoo    | 123 | 47.50     | 3.08    | 46.97      | 48.10       |

Table 1: Bill length summary by species (Palmer Penguins).

## 6 Supplementary Video

For a concise overview of uncertainty visualization design principles, see the public lecture: Building Effective Uncertainty Visualizations (StanCon 2020).

## 7 Reference List

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4. Hullman, J., Resnick, P., & Adar, E. (2015). *Hypothetical Outcome Plots Outperform Error Bars and Violin Plots for Inferences About Reliability of Variable Ordering*. <https://pmc.ncbi.nlm.nih.gov/articles/PMC4646698/>
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6. Palmer Penguins dataset (seaborn-data). <https://github.com/mwaskom/seaborn-data/blob/master/penguins.csv>
7. Kay, M. (2020). *Building Effective Uncertainty Visualizations* (StanCon lecture). [https://www.youtube.com/watch?v=wbzfqh\\_3LyM](https://www.youtube.com/watch?v=wbzfqh_3LyM)